

# Judgment and Prediction: From Experimentation to Delayed and Uncertain Experimentation (Uncorrected Proofs)

ECON 470 — Microeconomics of Artificial Intelligence

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## 1 Setup

An agent faces two equally likely states,  $\theta \in \{1, 2\}$ , with  $\Pr(\theta = 1) = \Pr(\theta = 2) = \frac{1}{2}$ . There are two actions:

- **Safe action** ( $a = 1$ ): pays  $S$  regardless of the state.
- **Risky action** ( $a = 2$ ): pays either  $R$  or  $r$  in each state, where  $R > S > r$ . The agent does not know which.

The agent does not know what the risky action pays in either state. **Judgment** is the process of experimenting — testing, piloting, researching — to find out. Each state requires its own experiment: learning what the risky action pays in state 1 (e.g., running a clinical trial for condition 1) tells you nothing about what it pays in state 2 (condition 2 requires its own trial). The probability that an experiment in a given state reveals a payoff of  $R$  is  $v$ , independently across states.

This gives three possible experimental outcomes when both states are investigated:

- Both states pay  $R$ : probability  $v^2$ .
- One state pays  $R$ , the other  $r$ : probability  $2v(1 - v)$ .
- Both states pay  $r$ : probability  $(1 - v)^2$ .

We assume:

- **A1 (Safe Default):**  $vR + (1 - v)r \leq S$ . Without experimentation, the safe action is optimal.
- **A2 (Judgment Insufficient):**  $\frac{1}{2}(R + r) \leq S$ . Even after learning that one state pays  $R$  and the other  $r$ , going risky in both states is suboptimal.

Parameters for all examples:  $R = 100$ ,  $r = 20$ ,  $S = 75$ ,  $v = 0.3$ .

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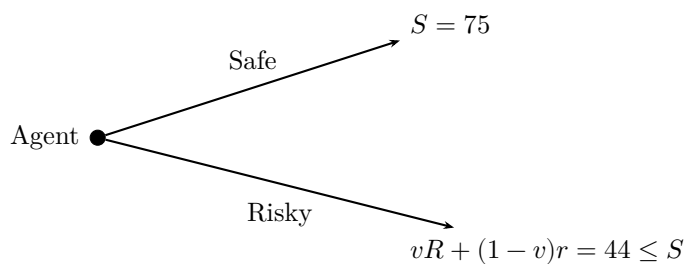
## Part I: Experimentation (Gans §5.1.1)

In this part, experimentation simply works: the agent learns the full payoff function — what the risky action pays in each state — with no delay and no chance of failure. This establishes the baseline complementarity result.

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## 2 Case 1: No Prediction, No Experimentation

The agent knows nothing about the state or the payoffs. By A1, the safe action is optimal.

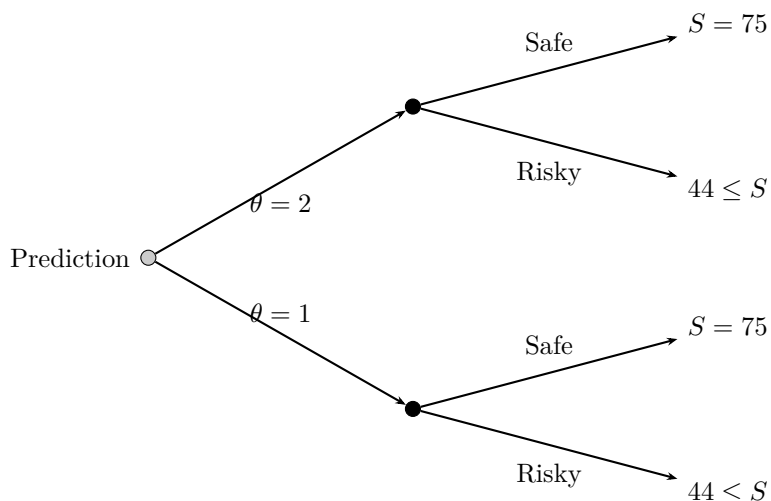


The agent chooses Safe. **Payoff:**  $V_\emptyset = 75$ .

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## 3 Case 2: Prediction Only (No Experimentation)

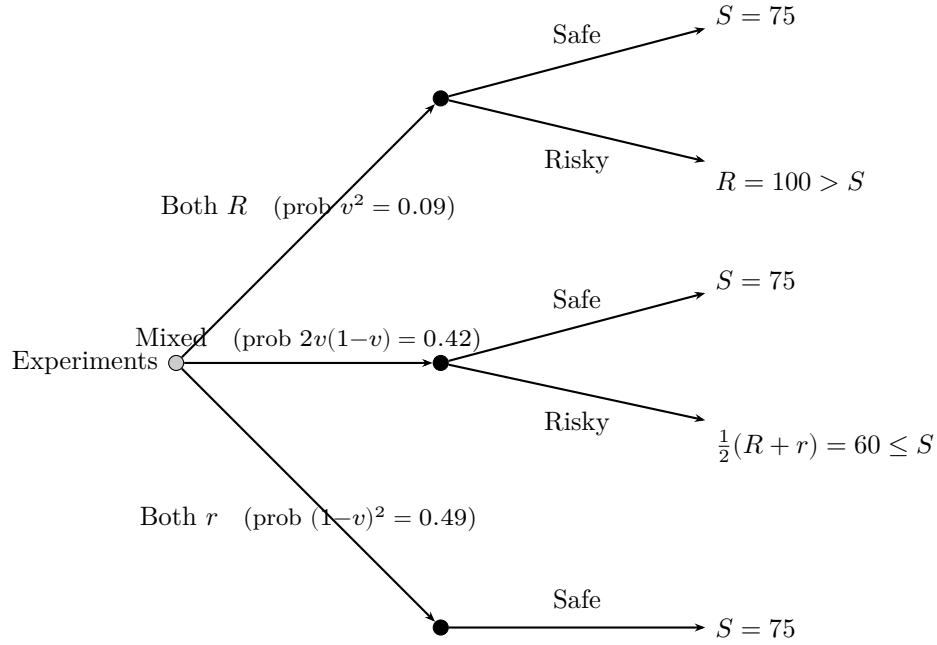
A prediction reveals  $\theta$ . The agent knows the state but has not experimented — they still don't know whether the risky action pays  $R$  or  $r$  in that state. By A1, the expected risky payoff remains  $vR + (1 - v)r = 44 < 75 = S$ .



Knowing the state does not help: the agent hasn't experimented, so they don't know what the risky action pays. Safe is optimal in both states. **Payoff:**  $V_P = 75$ .

#### 4 Case 3: Experimentation Only (No Prediction)

The agent experiments on both states and — with no delay and no chance of failure — learns the full payoff function. Three configurations are possible:



- **Both states pay  $R$**  (prob  $v^2 = 0.09$ ): risky dominates. Agent takes risky, earns  $R = 100$ .
- **Mixed** (prob  $2v(1-v) = 0.42$ ): one state pays  $R$ , the other  $r$ . The agent would like to go risky only in the  $R$ -state, but does not know  $\theta$ . Going risky in both gives  $\frac{1}{2}(R + r) = 60 \leq 75 = S$  by A2. Agent takes safe, earns  $S = 75$ . The experiment produced useful information — but it doesn't change the action.
- **Both states pay  $r$**  (prob  $(1-v)^2 = 0.49$ ): safe dominates. Agent takes safe, earns  $S = 75$ .

**Expected payoff:**

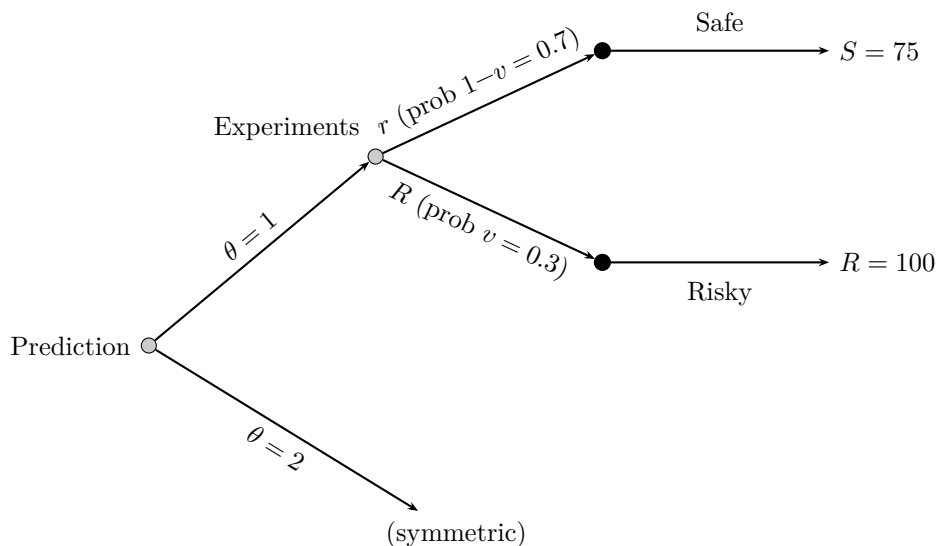
$$V_J = v^2 R + (1 - v^2) S = 0.09(100) + 0.91(75) = 9 + 68.25 = 77.25.$$

**Value of experimentation:**  $V_J - V_\emptyset = 77.25 - 75 = 2.25$ .

Experimentation is valuable — but only because of the  $v^2$  probability of both-states- $R$ . In the mixed case (probability 0.42), the agent learns that one state pays  $R$  but cannot target the risky action to that state without knowing  $\theta$ . The information is *inert* — it doesn't change behavior.

## 5 Case 4: Experimentation and Prediction Together

The agent knows  $\theta$  (prediction) and has experimented on both states. In the realized state, the agent knows whether risky pays  $R$  or  $r$  and acts accordingly.



- Risky pays  $R$  in the realized state (prob  $v = 0.3$ ): take risky, earn  $R = 100$ .
- Risky pays  $r$  in the realized state (prob  $1 - v = 0.7$ ): take safe, earn  $S = 75$ .

**Expected payoff:**

$$V_{PJ} = vR + (1 - v)S = 0.3(100) + 0.7(75) = 30 + 52.5 = 82.5.$$

## 6 Summary: Experimentation

|                      | No Experimentation | Experimentation  |
|----------------------|--------------------|------------------|
| <b>No Prediction</b> | $V_\emptyset = 75$ | $V_J = 77.25$    |
| <b>Prediction</b>    | $V_P = 75$         | $V_{PJ} = 82.50$ |

**Marginal value of prediction without experimentation:**  $V_P - V_\emptyset = 0$ .

**Marginal value of prediction with experimentation:**  $V_{PJ} - V_J = 82.50 - 77.25 = 5.25$ .

**Marginal value of experimentation without prediction:**  $V_J - V_\emptyset = 2.25$ .

**Marginal value of experimentation with prediction:**  $V_{PJ} - V_P = 7.50$ .

Both margins confirm **complementarity**: prediction is more valuable when the agent has experimented ( $5.25 > 0$ ), and experimentation is more valuable when the agent has a prediction ( $7.50 > 2.25$ ).

**The source of complementarity.** Without prediction, experimentation only changes the agent's action in the both- $R$  case (probability  $v^2 = 0.09$ ). The mixed case — where the agent learns that one state pays  $R$  and the other pays  $r$  — does not change the action because A2 makes going risky in both states suboptimal. Prediction resolves this: by revealing  $\theta$ , it makes every  $R$ -discovery actionable (probability  $v = 0.3$ , more than three times  $v^2 = 0.09$ ).

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## 7 What This Model Cannot Show

The experimentation model establishes that prediction and judgment are complements. But it cannot answer three important questions:

1. **How does the return to prediction vary with the quality of experimentation?** There is no quality parameter — experimentation either works or is absent.
2. **Is there a threshold below which experimentation is not worthwhile?** No — experimentation is always worth having ( $V_J > V_\emptyset$ ).
3. **Can we get the stronger result that neither alone has value?** No — experimentation alone gives  $V_J = 77.25 > 75 = V_\emptyset$ .

The next part introduces a two-period model where experimentation is delayed and may fail. This delivers all three results.

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## Part II: Delayed and Uncertain Experimentation (Two-Period Model)

We now relax both idealizations. Experimentation is delayed by one period (discount factor  $\delta < 1$ ) and is no longer guaranteed to work: it succeeds with probability  $\lambda \in (0, 1)$ . If it succeeds, the agent learns the full payoff function (what the risky action pays in each state). If it fails, the agent learns nothing. The agent has one attempt.

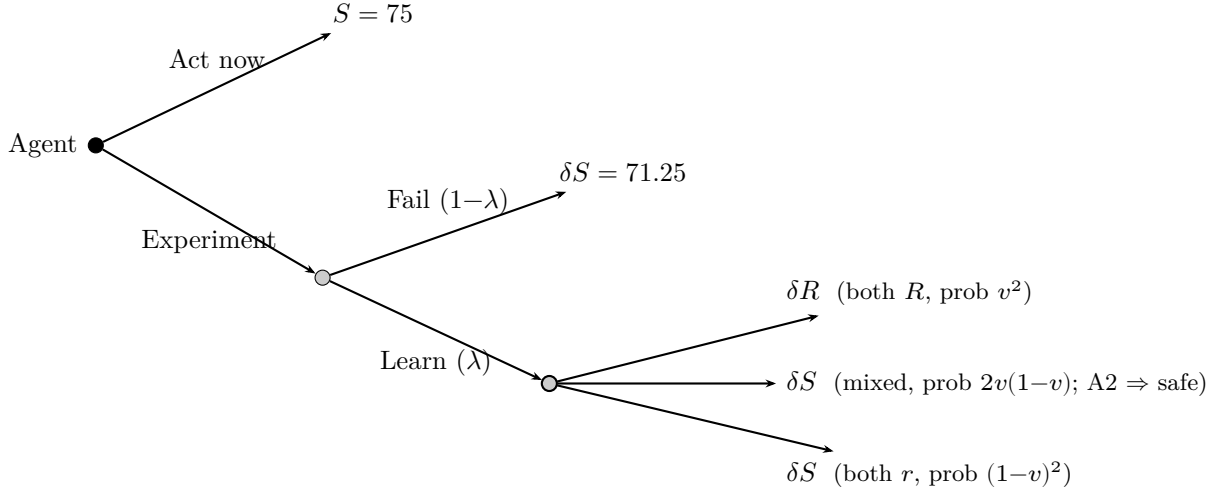
The parameter  $\lambda$  measures the **quality of experimentation** — how likely a single attempt is to resolve the agent's uncertainty about payoffs. Think of it as: how good is the agent's lab, testing process, or pilot program?

**Additional parameters:**  $\delta = 0.95$ .

## 8 Case 5: No Prediction, Delayed and Uncertain Experimentation

The agent can experiment (learn the payoff function with probability  $\lambda$ , at a one-period delay) or act now (earn  $S = 75$  immediately).

If the experiment succeeds, the analysis is identical to Case 3: only the both- $R$  configuration (prob  $v^2$ ) changes the action. But now the payoffs are discounted by  $\delta$ .



**Expected payoff from experimenting:**

$$V_{\text{exp}} = \lambda\delta[v^2R + (1 - v^2)S] + (1 - \lambda)\delta S = \delta S + \lambda\delta v^2(R - S).$$

**Value of experimentation** (relative to acting now and earning  $S$ ):

$$V_{\text{exp}} - S = \underbrace{\lambda\delta v^2(R - S)}_{\text{expected gain from both-}R} - \underbrace{S(1 - \delta)}_{\text{cost of waiting}}.$$

The first term is the probability of learning ( $\lambda$ ), times the discount ( $\delta$ ), times the probability that the experimental results are actionable ( $v^2$ ), times the payoff upgrade ( $R - S$ ). The second term is the impatience cost: by waiting one period, the agent loses  $S(1 - \delta)$  in present value even if they end up taking safe.

**With our parameters:**  $\lambda(0.95)(0.09)(25) = 2.1375\lambda$  vs  $75(0.05) = 3.75$ . The agent experiments iff  $\lambda > 1.75$ . Since  $\lambda \leq 1$ , the agent **never experiments** with these parameters.

This is not a general result — it depends on payoff parameters. Higher  $v$  (more likely that risky is good), higher  $R - S$  (bigger upside), or higher  $\delta$  (more patient) would all make experimentation worthwhile. But with our parameters, the actionable probability  $v^2 = 0.09$  is simply too small to justify the delay.

**Intuition:** The only configuration where experimentation changes the action is both- $R$  (probability  $v^2 = 0.09$ ) — too unlikely to justify the delay. In the mixed case, the agent learns that one state pays  $R$  and the other pays  $r$ , but without knowing  $\theta$ , they cannot target the risky action to the

$R$ -state. By A2, going risky in both is suboptimal. The information doesn't change the action. This is the information that prediction would make actionable.

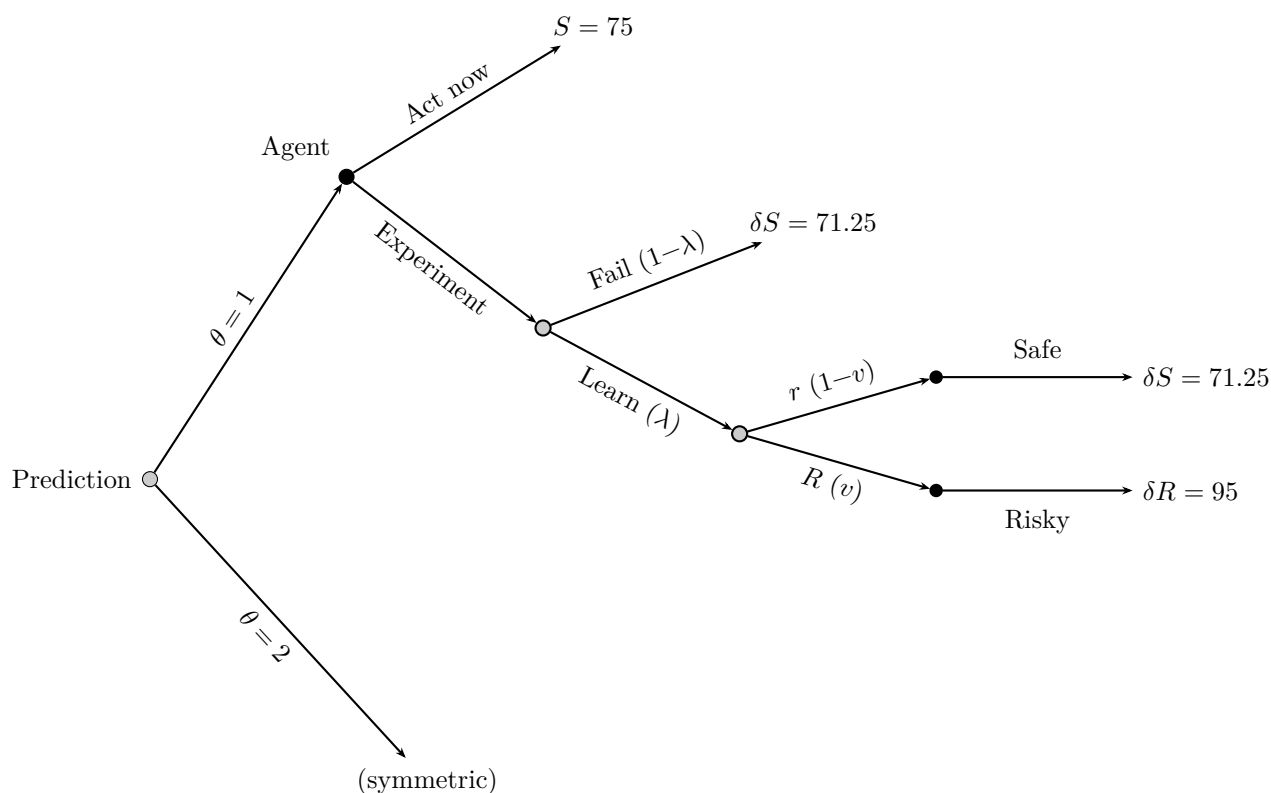
**Payoff:**  $V_J = S = 75$ .

## 9 Case 6: Prediction and Delayed, Uncertain Experimentation Together

The agent knows  $\theta$  (prediction) and can experiment to learn the payoff function (delayed and uncertain). The experiment still reveals payoffs in both states, but now the agent knows which state they are in — so they only need to act on the payoff in the realized state.

If the experiment succeeds:

- Risky pays  $R$  in the realized state (prob  $v$ ): take risky, earn  $\delta R$ .
- Risky pays  $r$  in the realized state (prob  $1 - v$ ): take safe, earn  $\delta S$ .



**Expected payoff from experimenting (given prediction):**

$$V_{PJ} = \lambda \delta [vR + (1 - v)S] + (1 - \lambda) \delta S = \delta [S + \lambda v(R - S)].$$

**Value of experimentation with prediction** (relative to acting now):

$$V_{PJ} - S = \underbrace{\lambda \delta v (R - S)}_{\text{expected gain from any } R} - \underbrace{S(1 - \delta)}_{\text{cost of waiting}} .$$

The structure is identical to Case 5, but with  $v$  instead of  $v^2$  in the gain term. The agent experiments iff:

$$\lambda > \frac{S(1 - \delta)}{\delta v (R - S)} \equiv \bar{\lambda} .$$

With our parameters:

$$\bar{\lambda} = \frac{75(0.05)}{0.95(0.3)(25)} = \frac{3.75}{7.125} \approx 0.53 .$$

For  $\lambda > 0.53$ , the agent experiments and earns  $V_{PJ} > 75$ .

**The comparison is sharp.** The value of experimentation has the same structure in both cases:

|                  | Without prediction (Case 5)                                 | With prediction (Case 6)                                  |
|------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| <b>Gain term</b> | $\lambda \delta v^2 (R - S)$                                | $\lambda \delta v (R - S)$                                |
| <b>Cost term</b> | $S(1 - \delta)$                                             | $S(1 - \delta)$                                           |
| <b>Threshold</b> | $\lambda > \frac{S(1 - \delta)}{\delta v^2 (R - S)} = 1.75$ | $\lambda > \frac{S(1 - \delta)}{\delta v (R - S)} = 0.53$ |

The cost of waiting is the same. The only difference is the actionable probability:  $v^2$  without prediction,  $v$  with prediction. Since  $v < 1$ , we have  $v > v^2$ , so prediction strictly lowers the threshold for experimentation to pay for itself. **Prediction makes it easier for experimentation to justify its cost.** That is complementarity.

## 10 Proposition

**Proposition.** Under A1 and A2:

1. Neither prediction alone nor experimentation alone has any value:  $V_P = V_J = V_\emptyset = S$ .
2. Prediction and experimentation together have positive value if and only if  $\lambda > \bar{\lambda}$ , in which case  $V_{PJ} = \delta[S + \lambda v(R - S)] > S$ .
3. The return to prediction is increasing in the quality of experimentation:  $\frac{\partial V_{PJ}}{\partial \lambda} = \delta v(R - S) > 0$ .
4. Prediction and experimentation (judgment) are **complements**.



**Proof.**

*Parts 1 and 2.* Established in Cases 1–2 (prediction alone), Case 5 (experimentation alone), and Case 6 (both together).

*Part 3.* When  $\lambda > \bar{\lambda}$ ,  $V_{PJ} = \delta[S + \lambda v(R - S)]$ . Differentiating:  $\frac{\partial V_{PJ}}{\partial \lambda} = \delta v(R - S) > 0$  since  $R > S$  and  $\delta, v > 0$ . Better experimentation (higher  $\lambda$ ) increases the return to having a prediction.

*Part 4.* The marginal value of prediction without experimentation is  $V_P - V_\emptyset = 0$ . The marginal value of prediction with experimentation is  $V_{PJ} - V_J > 0$  when  $\lambda > \bar{\lambda}$ . Since a positive number exceeds zero, prediction is strictly more valuable when experimentation is available.  $\square$

**The mechanism.** Without prediction, experimentation reveals the full payoff function but the agent’s action changes only when the risky action dominates in *all* states (probability  $v^2$ ). In the mixed case — one state pays  $R$ , the other  $r$  — the agent knows the payoffs but cannot target the risky action to the  $R$ -state without knowing  $\theta$ . Going risky in both states is suboptimal by A2, so the agent takes safe and the experimental results do not change the action. Prediction resolves this: by revealing  $\theta$ , it lets the agent go risky in exactly the state that pays  $R$ . Every  $R$ -discovery (probability  $v$ ) becomes action-changing, not just the both- $R$  coincidence (probability  $v^2$ ). Prediction doesn’t change what the agent learns; it changes *how much of what they learn changes their action*.

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## 11 Numerical Example

| $\lambda$ | Experiments?                          | $V_{PJ}$                  | Marginal value of prediction |
|-----------|---------------------------------------|---------------------------|------------------------------|
| 0.4       | No ( $< \bar{\lambda} \approx 0.53$ ) | 75.00                     | 0.00                         |
| 0.7       | Yes                                   | $0.95[75 + 5.25] = 76.24$ | 1.24                         |
| 0.9       | Yes                                   | $0.95[75 + 6.75] = 77.66$ | 2.66                         |

The marginal value of prediction more than doubles as  $\lambda$  rises from 0.7 to 0.9. Better experimentation makes prediction more valuable — the signature of complementarity.

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## 12 Comparison to Gans §5.1.2

|                                     | Gans §5.1.2                                           | Two-period model     |
|-------------------------------------|-------------------------------------------------------|----------------------|
| <b>Time horizon</b>                 | Infinite, geometric discounting                       | Two periods          |
| <b>What experimentation reveals</b> | Payoff in one state per attempt                       | Full payoff function |
| <b>Quality parameter</b>            | $\hat{\lambda} = \frac{\lambda}{1-(1-\lambda)\delta}$ | $\lambda$ directly   |

|                                  | Gans §5.1.2                                                                | Two-period model                                                                       |
|----------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Number of attempts</b>        | Unlimited (agent keeps trying)                                             | One shot                                                                               |
| <b>Cost of experimentation</b>   | Delay (opportunity cost of $S$ )                                           | Same                                                                                   |
| <b>Key result</b>                | Return to prediction<br>increasing in $\hat{\lambda}$                      | Return to prediction<br>increasing in $\lambda$                                        |
| <b>Source of complementarity</b> | Prediction <i>focuses</i><br>experimentation (one state<br>instead of two) | Prediction makes experimental<br>results <i>actionable</i> ( $v$ instead of<br>$v^2$ ) |

In Gans’s model, the agent experiments on states one at a time and has unlimited attempts, so experimentation alone *can* be valuable — the agent may learn enough to act even without knowing the state. Prediction helps by *economizing* on experimentation: the agent only needs to investigate one state instead of two.

In the two-period model, the agent learns the full payoff function in one shot but has only one chance. Without prediction, the mixed-case results (one state pays  $R$ , the other  $r$ ) do not change the agent’s action — going risky in both states is suboptimal by A2, so the agent takes safe regardless. Prediction changes this: knowing  $\theta$  lets the agent go risky in exactly the  $R$ -state, turning experimental results that were previously *inert* into results that change behavior.

The core insight is identical: **prediction tells you where you are; experimentation tells you what to do there. Neither is useful alone. Together, they unlock decisions that neither could support on its own.**